Project Record - part 7 of 8

Utensil Theories Transfer Protocol, 17 to 21 October 2026. This part holds: 8. What already exists in research, and where the owner's credit could come from; 9. Where things stand now, and what comes next; 10. Every file and where it lives.

8. What already exists in research, and where the owner's credit could come from

Sources: `experiments/PRIOR_WORK.md` (18 works, each confirmed by a separate skeptic agent) and `GROW_UNDERSTANDING_2026-10-20/reports/fact_check_and_credit.md` (21 facts, checked through web-search extracts because direct page fetches were blocked).

Already published: stories, morals and marker words

The owner's idea	Closest existing work	How close
Stories teach values better than rules	Riedl and Harrison (AAAI-16 workshop)	Same big idea; no controlled test
Learn norms from stories, then transfer	Nahian, Frazier, Riedl and Harrison (AIES 2020)	Close
Fable-to-moral matching	BIG-bench understanding_fables; MORABLES (2025); STORAL (2022); ePiC (2022)	The test is not new
Judging right and wrong acts	Moral Stories (2021); ETHICS (2021)	Standard benchmarks
Value-marker words in training text	Korbak et al. (2023); CTRL (2019); Quark (2022)	Established, but each tag stands alone, with no carry
A running balance used up as text goes on	Decision Transformer (2021); Kikuchi et al. (2016)	Same mechanism, for rewards or length
Training reading shapes values	Tice et al. (2026); Anthropic, "Teaching Claude why" (2026)	Supports the direction

Already published: compression and seeds

- **Understanding = prediction = compression:**
  - Shannon (1948/1951);
  - Kolmogorov complexity, the idea that information is the length of the shortest program that makes the data (Solomonoff 1960/64, Kolmogorov 1965, Chaitin 1966/69);
  - the Hutter Prize (2006 onward);
  - Delétang et al. 2023;
  - Huang et al. 2024 (r = -0.93 between compression and benchmark scores).
- **Seed plus shared model regrowing exact data:**
  - Schmidhuber and Heil 1996, cmix, NNCP, LLMZip (2023) and ts_zip.
  - The 2026 Hutter record, fx2-cmix-T (100,424,672 bytes), uses a small pre-trained model as a shared greenhouse. It was seen only through search extracts, since it is after the AI's training data.
- **Shared libraries:** Brotli's built-in 122,784-byte dictionary, zstd dictionaries, and Rinberg et al. 2026.
- **Other known results:**
  - Data on a fixed page: the pi filesystem joke.
  - The counting argument.
  - Determinism as a requirement (the palm-seed problem).
  - Marker tokens: CTRL 2019 and MeCo 2025.

What could still be new ("not found" is not proof)

1. **A measured result for the owner's training signal.** A carried right/wrong balance that makes a small model a better judge, or a better exact compressor, of unseen stories, with fair controls. The old shifting word did not do this on compression. The "17 + 4" recharge now in the numbers test is the next candidate. An honest negative result is also publishable.
2. **The shifting word itself**, and a controlled stories-vs-rules test trained from scratch.
3. **"Seed bits as an understanding meter":** measure the bits needed to regrow a story exactly, before and after a lesson. The drop is understanding gained.

Authorship credit vs discovery credit

- **Discovery credit** needs a new measured finding. "Compression = understanding", exact regrowth from a seed, and any method that shrinks all data are not claimable: they are published or impossible.
- **Authorship credit** covers original explanation: the greenhouse story and the AI-free seed test as a teaching demo.
- **The repository is public under MIT** (an open license that lets anyone copy and reuse it). Anyone may reuse the code and ideas, and MIT does not protect ideas. The timestamped commits show when each result was published.

Outside AI reports, and how they were checked

18 October: a two-version "Data Filtering & Methodology Report". The owner pasted it in for evaluation. The session record does not name which AI wrote it. The AI found it was a summary of the project's own write-ups, with no independent analysis.

- **Right:** Aesop at chance, the carry idea helped only in long stories, and 0 of 15 acts flipped without the rule verses (later corrected to 0 of 14, since one act was too rare to measure).
- **Overstated:** "demonstrated value" should read "supported once", and "identified the core defect" was only a likely explanation.
- **Wrong:** it proposed a reset that already exists, mixed up the sub-word test, and cited only the written documents while claiming to have analysed the raw results.
- **Its biggest gap:** drift away from the communication goal.

Version 2 was much better but still had three errors and two gaps. The AI wrote correction lines for the owner to paste back.

20 October: the owner's 98.9% chat results. The AI explained how chat tests can look better than they are: the chat sees the original, and size can hide in the grow instructions ("huge files") or in long numbers. It then built the fair seed test. So far, 98.9%-level reductions appear only on patterned data. The 98.6% library result is the likely, unproven explanation for the chat results.

9. Where things stand now, and what comes next

Status at the time of this record (21 October 2026).

9.1 Every task

#	Task	Status
1	Confirmation on 100 new stories (exp 8)	Done
2	Aesop test review fixes	Done
3	Aesop moral transfer test (exp 9)	Done: not confirmed
4	Handoff files for a new session	Done; kept current
5	Method comparison (exp 10)	Done: a tie
6	Long-story carry test (exp 11)	Done: supported once
7	Audit corrections, file delivery, CLAUDE.md	Done
8	Numbers test	Running: full run started 17:45 UTC, 21 Oct; no results yet
9	The "2" test	Queued, owner-approved

#	Task	Status
10	The +/- agent	Queued, owner-approved
11	This project record	In progress
-	Owner's comprehension test across AIs	On the owner's side

9.2 The numbers test right now

Update, 21 October, 17:45 UTC: the review finished.

- Three of four reviewers said "ready", with no blocking problems and 12 minor notes.
- The run-safety checks were done directly after the fourth reviewer was cut off twice by the usage limit.
- Ten small fixes leave the frozen design unchanged. Among them, the inputs are now pinned: the run refuses to start if the word values or any of the six markings differ from the verified versions.
- A second dry run passed, and **the full run started at 17:45 UTC.**
- Every finding is in `NUMBERS_TEST_2026-10-21/review/review_findings_and_fixes.md`.
- **What it tests:** the owner's numbers 17, 32, 16, 34, 47 in the charge slot of "W1 = 17 + 4 Remaining". This is the first test of the recharge rule. Six versions (`num_charge17`, `num_seq`, `num_seqrev`, `num_const29`, `num_charge47`, `num_drain0`), 3 seeds each: 18 tiny models. Pass/fail rules were written before training.
- **Dry run passed** on 21 Oct, 02:06 to 02:08 UTC (1.7 minutes by its own log, 5-step models, scratch folder only). All six pass the manipulation check (at most 1.23% of verses at the cap; every spelling on at least 3% of verses). The stored control scores were reproduced exactly. Every exact seed regrew byte for byte. The dry-run numbers mean nothing about the result.
- **Independent review:** four reviewers (rule arithmetic, marker scoring, statistics, long-run safety), with a separate checker for each serious finding. The fixes were made before the full run.
- **Full run:** about 4 to 5 hours, saved and pushed every 10 minutes (`NUMBERS_TEST_2026-10-21/code/run_full.sh`, `autosave.sh`).
- **NO RESULTS YET.** At the time of writing the `results/` folder is empty, and `logs/` holds only the full-run log, which shows the first model has started training.
- **The main verdict** is judged on 100 story pairs. "The owner's numbers matter" means `num_seq` beats `num_charge17`. "The recharge rule helps" means `num_charge17` beats the current shifting word and also beats `num_drain0`. Anything else is "not shown at this scale", never "disproved". The written expectation (not part of the verdict) is a tie.

9.3 Queued next, owner-approved

1. **The "2" test.** Write the Remaining value into the text as a number the model can read. Today's tokenizer drops digits.
2. **The +/- agent.** Classify every input as + or - information, keep a running balance with the owner's rule, and report only its placement on the line (far left to far right), never the exact number. The owner: "dont do this step yourself but make sure it happens." The owner's scoring rule (21 Oct, in `HANDOFF.md`): many small negatives, fewer but larger positives, every input pulls a little negative, and over ordinary text the two sides balance. Not yet tested. Today's word values are symmetric (36.0% negative vs 37.4% positive valued words; average sizes 2.31 vs 2.27).
3. **The owner's comprehension test across AIs.** The same formula prompt and a shuffled control, given to Gemini (memory on and off), Copilot and Claude. So far, turning Gemini's memory off changed its whole answer, and neither Gemini nor Copilot said the shuffled control had no planned meaning. The owner is bringing back more answers.

9.4 Open threads (from `HANDOFF.md`, "Carried balance: open threads")

Thread	Lean so far	What would settle it
Does labeling teach judging?	Strongly for: about 69 vs 50 per 100, 3 runs; a side measure	A pre-registered judging test on fresh stories

Thread	Lean so far	What would settle it
Does the carry add anything beyond labeling?	Supported once, long stories only (+6 per 100, about 65 ties per 100)	A fresh long-story set at several distances
Why does it misjudge explicit verdicts?	Its marker updates too little	Check whether greedy marker choice hides updates; then retrain
Predicting consequences?	Against: 51/100	A bigger model, or matched ending formats
Stories vs rules?	Unclear	Aesop test was not confirmed
Re-adding 17 ("W1 = 17 + 4")	Untested	The numbers test, now
Do 32, 16, 34, 47 matter?	No reason yet	The numbers test, now
Do shared syllables matter?	Untestable with word tokens	A sub-word tokenizer test
A large AI as reader	Not run: needs an API key	The owner adds a key

9.5 Parts of the owner's idea still untested

- The recharge rule and the numbers (results pending).
- The number form "2" and the zero symbols ?0, -0, +0, 00.
- The owner's uneven +/- scoring and the +/- agent.
- Shared syllables (needs a sub-word tokenizer).
- Models trained on stories and scored on held-out stories.
- Anything on a large AI or in a real conversation (PLAN step 1b: the conversation-carry test and the owner-explanation test).
- The owner's own graph factors behind "strawberry, orange, cucumber".

10. Every file and where it lives

- **Repository:** the owner's public GitHub repository (MIT license), branch `ccr-1be0d913-1tlzao`. **These files are not on `main`.**
- **Out of scope:** `src/`, `docs/`, the book PDFs, `.doc` and `.html` files at the top level belong to the owner's separate zine project.
- **Commits:** 122 commits on this work, from 17 October to 17:48 UTC on 21 October (`git log` on the branch).

10.1 Delivery folders (top level of the repository)

Folder	Zip next to it	What is inside	Size
`AUDIT_PACK_2026-10-18`	`.zip`, 1.8 MB	Master audit guide (.docx); prompt for another AI; research notes, corrected and as audited; all code; raw results incl. flawed runs; a zine file (original web page named .doc, plus a real .docx); number-check script; strict file rules; build script; a copy of the Aesop folder	3.9 MB
`AESOP_RESULTS_2026-10-18`	`.zip`, 0.76 MB	Start-here guide and write-up (.docx and .md), raw results, code	1.6 MB
`TECHNICAL_REPORT_2026-10-20`	`.zip`, 0.47 MB	Seed-and-grow rebuild kit as real .doc (Word 97-2003), .docx and .md (full version 259 KB); copies of all code; reference code never run	1.3 MB
`SEED_TEST_2026-10-20`	`.zip`, 0.1 MB	`seed_test.py`; three graph methods in `methods/`; checker reports in `checks/graph_methods_reports.md`; results; test inputs	0.3 MB
`GROW_UNDERSTANDING_2026-10-20`	`.zip`, 0.18 MB, plus 3 model zips	Experiment 12 code, results, three reports (`reports/`), the 15 models, `SHA256SUMS.txt`, the Bible text	202 MB

Folder	Zip next to it	What is inside	Size
NUMBERS_TEST_2026-10-21	none yet	Start-here guide, code/ (script, launcher, auto-saver, Word converter), design/ (readings, judge's design, simulation), dry_run/, review/ (review findings and fixes), logs/ (the full-run log); results/ empty at the time of writing	0.65 MB
PROJECT_RECORD_2026-10-21	to come	This record and its build scripts (code/assemble_report.py, md2pdf.py, build_outputs.py)	in progress

Each folder has a 00_START_HERE guide listing every file, its path and where the original lives.

10.2 Key code and results in experiments/

- **Write-ups and notes:** README.md (experiments 1 to 12), HANDOFF.md, PLAN.md, COMMUNICATION_GUIDE.md, PRIOR_WORK.md, NEXT_SESSION_PROMPT.md.
- **Code:** bible_understanding.py (exp 1-4), consequences_test.py (5), tiny_transformer.py (6, and the model all later tests use), shifting_word.py (7), stories_v2.py and confirm.py (8), aesop_test.py and world_stories.py (9), method_comparison.py (10), long_story_test.py (11), understanding_compression.py (12), numbers_test.py (the numbers test, 65 KB).
- **Results:** results.txt, results_consequences.txt, results_transformer.json, results_shifting_word.json, results_confirm.json, results_methods.json, results_long_story.json, the two flawed long-story runs, results_long_story_startmarker.json, results_aesop.json (1.5 MB) and .txt, results_understanding.json (441 KB) and .txt.
- **First script:** resonance/ (engine, 16 tests, README).

10.3 The models: what is in git and what is not

Models	Count	Size	In git?
Control greenhouses: baseline, scrambled, shifting, label, carry3, seeds 0-2 (experiments/models/ and the GROW folder)	15	about 13.8 MB each, 198 MB in all	Yes, force-added (saved despite the repository's rule to skip model files); git stores identical copies once
The same 15, zipped by seed (GROW_UNDERSTANDING_2026-10-20_models_seed0/1/2.zip)	3 zips	about 63.8 MB each	Yes
Aesop test models (aesop3_*)	16	235 MB	No: local only, retrain with aesop_test.py
Numbers test models (num_*)	18	not trained yet	Not decided; the auto-saver commits results only

- **Why the zips are split:** GitHub refuses any single file over 100 MB, so the 15 models went into three zips of 5.
- **Data:** experiments/data/KJV.txt (4.2 MB) is in git. The world stories (3.8 MB) and novels (12 MB) are not; they re-download from a GitHub mirror.
- **Repository history:** about 386 MB.